Section 4: Case Study – Getting To Work

Relates to Questions 40-52

20 Marks available in this Section

INTRODUCTION

For this case study, you have just moved to a new city for a new job on a 3-year contract. You are considering five different possible options to get to work - cycling, bus, train, petrol vehicle and electric vehicle.

In order to work out the best transport method to use, you decide to prepare a budget of the estimated cost of each of these options. Your budget will be monthly over 3 years (36 months from 1 January 2019 until 31 December 2021). You do not need to consider inflation, financing costs or the time value of money in your budgeting.

The assumptions below are also provided in the relevant case study workbook.

GENERAL ASSUMPTIONS

All travel time is costed at $20 per hour.

Your work the following number of days per month:

January: 29 days
February: 27 days
March: 30 days
April: 11 days
May: 9 days
June: 10 days
July: 16 days
August: 19 days
September: 18 days
October: 26 days
November: 24 days
December: 21 days

CYCLING ASSUMPTIONS

Cycle purchase (One-off in January 2019): $500. No residual value at the end of the budget period.

Cycle maintenance (Semi-annually, first in July 2019): $50.

Travel time (Per trip, one way): 60 minutes.

BUS ASSUMPTIONS

Bus fare (Per trip, one way): $5.

Travel time (Per trip, one way): 40 minutes.

TRAIN ASSUMPTIONS

Train pass (Quarterly, first in January 2019): $540.

Travel time (Per trip, one way): 35 minutes.

CAR ASSUMPTIONS - GENERAL (APPLY TO BOTH PETROL AND ELECTRIC VEHICLE OPTIONS)

Distance to drive (Per trip, one way): 14km.

Average speed: 50km per hour.

Parking (Daily): $10.

Your travel time for driving should be based on the distance and average speed only - parking time is immaterial.

CAR ASSUMPTIONS - PETROL VEHICLE

Petrol car purchase (One-off in January 2019): $6,000.

Expected residual petrol car value (One-off in December 2021): $2,000. Treat this as a negative cost.

Petrol (fuel): 42 cents per km.

Petrol car maintenance (Semi-annually, first in July 2019): $200.

Insurance (Annually, first in January 2019): $300.

CAR ASSUMPTIONS - ELECTRIC VEHICLE

Electric car lease (Monthly): $445.

Maintenance and insurance is included in the lease.

Free charging electricity is available so no fuel cost, parking cost required.

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Round 1 Section 4 - Case Study Information Pack

Prepare your model and then use it to answer the given questions.

When finished, please upload your workbook (Question 52).

QUESTIONS

Question 40

What is the total time cost for cycling over the first 6 months? [1 mark]

A. 4,520

B. 4,540

C. 4,560

D. 4,580

E. 4,600

F. 4,620

G. 4,640

H. 4,660

I. 4,680

Question 41

What is the total cost for cycling over the first 9 months? [1 mark]

A. 7,300

B. 7,310

C. 7,320

D. 7,330

E. 7,340

F. 7,350

G. 7,360

H. 7,370

I. 7,380

Question 42

What is the total bus fare paid from March 2019 until November 2019? [1 mark]

A. 1,570

B. 1,580

C. 1,590

D. 1,600

E. 1,610

F. 1,620

G. 1,630

H. 1,640

I. 1,650

Question 43

What is the total cost for the train between May 2020 and December 2020? [1 mark]

A. 4,413

B. 4,414

C. 4,415

D. 4,416

E. 4,417

F. 4,418

G. 4,419

H. 4,420

I. 4,421

Question 44

What is the total cost, excluding time, for the petrol vehicle over the three year period? [2 marks]

A. 21,564

B. 21,565

C. 21,566

D. 21,567

E. 21,568

F. 21,569

G. 21,570

H. 21,571

I. 21,572

Question 45

What is the total cost electric vehicle over the three year period? [2 marks]

A. 31,276

B. 31,277

C. 31,278

D. 31,279

E. 31,280

F. 31,281

G. 31,282

H. 31,283

I. 31,284

Question 46

Based on the total cumulative cost over the three years, excluding the time cost, which is the most expensive option? [2 marks]

A. Cycling

B. Bus

C. Train

D. Petrol vehicle

E. Electric vehicle

Question 47

Based on the total cumulative cost over the three years, which is the cheapest option? [2 marks]

A. Cycling

B. Bus

C. Train

D. Petrol vehicle

E. Electric vehicle

Question 48

In how many months is the cumulative bus cost (excluding time) less than the cumulative train cost (excluding time)? [2 marks]

Please enter your answer as a whole digit (e.g. 3)

Question 49

You decide to consider alternating between train and bus based on whichever is cheaper for the relevant quarter (including time cost). Under this approach, how many months will you take the bus? [2 marks]

A. 0

B. 3

C. 6

D. 9

E. 12

F. 15

G. 18

H. 21

I. 24

Question 50

For this question only, you decide to value your cycling time at $15 per hour, reflecting the exercise benefits. What is the revised total cycling cost over the three year period? [2 marks]

Please enter your answer to the nearest dollar (e.g. $1,000)

Question 51

You learn that there are transport upgrades planned that will impact on the travel time for the bus, train and car transport options.

The upgrade works will take place between October 2020 and March 2021 (inclusive). During this period, train travel time will increase by 30%, bus travel time will increase by 10% and car travel time will increase by 15%.

From April 2021 the upgrade works will cease. From this point onwards train travel time will decrease by 40%, bus travel time will decrease by 20% and car travel time will decrease by 5% (all decreases relative to the travel times before the upgrades took place).

Based on this, what is the total time cost over the three years of all five transport options combined? [2 marks]

Please enter your answer to the nearest dollar (e.g. $1,000)

Question 52

Please upload your workbook for this section.